Biltong — Salt, Vinegar, Six Days of Moving Air

Never cooked: the overnight vinegar bath and a controlled hang are the kill step, the salt only seasons. Vinegar well past 3% of the batch, at least six days on the hook, done by weight.

Yield: **12 × 125g** (from 3 kg raw) Active: **40 min** Total: **7 days** Box: **20–26°C** (68–79°F) · **45–60% RH** Done: **≥6 days + 50% loss**

WHERE IT WAS BORN
Southern Africa. Dutch *bil* (rump) + *tong* (strip). Said to begin with salt-and-sun drying older than the settlers, who allegedly added vinegar and Cape spices; the Voortrekkers dried it up to two weeks for the wagons.

EATEN, AUTHENTICALLY
In the hand: a whole stick, cut across the grain with a pocket knife as it is eaten. Beef or game; salt, vinegar and coriander the constants.

HOW IT'S USED NOW
A boxed, fan-dried snack sold wet, medium or dry by weight lost. Lab-validated in 2020: 5-log *Salmonella* kill with no heat, in 6–8 days.

INGREDIENTS

WEIGHT (G) · CUPS / VOLUME IN RIGHT COLUMN

THE MEAT — ABOUT 35 STRIPS

Beef silverside *(or topside, trimmed weight)*	3000g	

THE DRY CURE

Coarse sea salt *(2.2%)*	66g	3⅔ tbsp
Coriander seed *(whole; toast, crack)*	45g	7 tbsp
Dark brown sugar	30g	2 tbsp
Black pepper *(coarsely cracked)*	15g	2 tbsp
Bicarbonate of soda *(optional)*	3g	⅔ tsp
Curing salt #1, 6.25% *(optional; 0.01 g scale)*	5.7g	weigh it

THE BATH

Malt or cider vinegar *(5% acidity)*	450ml	1⅞ cups
Worcestershire *(contains fish)*	90ml	6 tbsp

THE RATIO — TRIMMED RAW BEEF = 100%
Salt 2.2% · Vinegar 15% · Coriander 1.5% · Sugar 1% · Pepper 0.5% · Bicarb 0.1% · Cure #1 0.19%. Salt 1.7–2.7% made no difference to the kill; vinegar at 2% failed, 3–4% passed (Karolenko 2020). 0.19% cure = 79 mg/kg nitrite ion, under the EU 80.

Kit: 0.01 g scale for the cure · non-reactive trays · stainless S-hooks · biltong box or cabin with a small fan · hygrometer · kitchen scale for the test strip · marker

METHOD

1. **Cut with the grain.** Firm 30–45 min in the freezer, silverskin off. Strips **2–2.5 cm** thick, 4–5 cm wide, 20–30 cm long. Same thickness throughout: the thickest sets the date.
2. **Toast, crack, mix.** Dry-toast the coriander until fragrant, crack coarsely. Mix with salt, sugar, pepper, bicarb and the weighed cure.
   > *EU since 9 Oct 2025: 80 mg/kg nitrite ion ingoing in non-heat-treated meat. The packet's 2.5 g/kg = 105 — over. 1.9 g/kg = 79. — Reg. (EU) 2023/2108*
3. **Season, rest.** Press the cure into every face. Cover, **1 h at ≤5°C**.
4. **Vinegar bath, overnight.** Vinegar + Worcestershire over, turn to coat, cover. **12–18 h at ≤5°C**, turned once.
   > *Validated: 16–20 h at 5°C in 10% vinegar → 5-log kill after 6 days' drying. 2% vinegar failed. — Karolenko et al., Microorganisms 2020*
5. **Drain, weigh, hang.** Pat, never rinse; re-dust coriander. Mark a test strip, write its day-0 weight. S-hooks, 3–4 cm apart, touching nothing.
6. **Dry ≥6 days.** 20–26°C, 45–60% RH, fan always on. Never down before day 6, whatever the scale says.
   > *Too hot, too fast or too dry seals a crust over a wet centre (case-hardening). At the end, surface a_w 0.80, inside 0.89–0.90. — Jones 2017; Karolenko 2020*
7. **Read it by weight.** Loss = (day 0 − today) ÷ day 0. ~35% wet (fridge, eat in a week) · **~50% medium** · 60%+ dry.

Done: test strip half its day-0 weight, at least six days hung; outside dark and dry, cut face deep red and firm right through, no wet sheen; medium bends before it snaps.
Serve: keep sticks whole; slice to order, thin, across the grain.

ONE CURE, THREE BATCHES
Wagyu 500 g, BMS 4–6: coriander light, cider vinegar, cure recommended (antioxidant for the fat); full bath, full 6 days, stop at 40–45% loss; fridge, eat within 2 weeks. **Dock sale** 10 kg: bath fixed at 16 h, cure every batch, release at ≥6 days *and* ≥50%, logged; register first (Reg. 852/2004), ask the local authority before the first bag.

ELEVATION

TIER 1 — NO EXTRA TIME
- **Malt vs cider** — malt for the classic sour, cider for a cleaner one
- **Chilli** — +3 g flakes per kg in the cure

TIER 2 — WORTH IT
- **Smoke** — +5 g smoked paprika per kg
- **Game** — kudu, springbok or venison loin for the beef

TIER 3 — RESTAURANT
- **Wagyu** — sliced paper-thin, laid out as charcuterie
- **Two tiers** — guests the wagyu, crew the silverside, one cure

TROUBLESHOOT

Hard out, wet in	Case-hardening → cooler, slower fan, more humidity
Stalls at 40%	Room too humid → dehumidifier or A/C
White bloom	Dry, wipes off with vinegar: salt. Fuzzy or coloured: mould, discard
Too salty	Salt concentrates as water leaves → take down at 50%

CHARTER PREP & STORAGE

At sea	**Dry in A/C or by a dehumidifier: air wetter than the stick puts water back**
Keep	Whole sticks, paper bags, cool and dry; vacuum-pack and freeze for longer
Fridge	Wet biltong and wagyu, always
Never	Less vinegar, a hot oven or the engine room, down before day 6

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Science: Karolenko et al., Microorganisms 2020 · Burnham et al., J. Food Safety 2008 · Jones et al., S. Afr. J. Anim. Sci. 2017 · Reg. (EU) 2023/2108